

Deepak Mallampati

APPLIED AI ENGINEER - CLINICAL AI AGENTS & RAG

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US-based · F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI Engineer with 4+ years building production Generative AI and LLM systems for regulated, high-stakes domains - including **healthcare RAG**, agentic orchestration, and clinical-data privacy. Ships reliable clinical-grade AI: hybrid retrieval with cross-encoder re-ranking to reduce hallucination, conductor-subagent agent orchestration with human-in-the-loop safety, and RAGAS/LLM-as-judge evaluation wired to dashboards. Deep experience with HIPAA workflows, differential-privacy on clinical datasets, and QLoRA fine-tuning - well-suited to building production clinical AI agents where reliability directly affects patient outcomes.

CORE COMPETENCIES & SKILLS

Clinical AI Agents · RAG & Retrieval · Agent Orchestration · LLM Evaluation (RAGAS / LLM-as-judge) · HITL Safety Guardrails · HIPAA & Clinical Data · Fine-tuning (QLoRA) · Observability & Cost Efficiency

Generative AI & LLMs: GPT-4o/GPT-5, Claude Sonnet, Gemini, LLaMA · LangChain/LangGraph · RAG Architecture · Prompt Engineering · Multi-Agent Systems · Fine-tuning (LoRA, QLoRA, PEFT) · Content Safety · Hugging Face Transformers

Vector Search & Retrieval: FAISS, Pinecone, Weaviate · Semantic & Hybrid Search · Cross-Encoder Re-ranking · Document Chunking · Semantic Caching

ML & NLP: PyTorch, TensorFlow, Keras · NER, Classification, Summarization · A/B Testing · Drift Detection · RAGAS Evaluation · BERTScore

Security & Compliance: HIPAA, SOC 2, PCI DSS · PII Masking/Redaction · Prompt Injection Mitigation · Audit Logging

MLOps & Cloud: AWS (Bedrock, SageMaker, Lambda, S3), Azure (OpenAI, AI Services, Content Safety), GCP Vertex AI · Docker, Kubernetes · CI/CD · MLflow, W&B · Model Monitoring & Observability

Languages & Tools: Python, JavaScript/TypeScript, Java, SQL · FastAPI, React · Git, Jira · Agile/Scrum

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present

AI Engineer

USA

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents - reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale healthcare document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination on domain-specific queries.
- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency/accuracy dashboards to benchmark iterations pre-deployment and surface regressions before release.
- Designed **privacy-preserving data workflows** for sensitive clinical datasets (K-anonymity, L-diversity, differential privacy) to enable compliant analytics with zero PII exposure.
- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry logic, reducing failed production runs by **60%**.
- Optimized inference cost and latency through prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.

Vanguard

Jan 2024 - Jan 2025

AI Engineer Intern

USA

- Developed **AI agents and backend services** for Vanguard's internal AI platform across **25+ AI agents** and workflows, integrating with production data pipelines and observability tooling.
- Engineered safeguards and content controls for secure, policy-compliant AI responses, reducing unsafe outputs by **42%** while maintaining response quality.
- Optimized prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by **27%** and informing platform model-selection decisions.
- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by **40%**.
- Partnered with product, platform, and data teams to deliver scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams.

Kent State University

Oct 2023 - Jan 2024

ML & Gen AI Research Assistant

USA

- Built a **privacy-preserving ML pipeline** on a clinical hospital-readmission dataset (k-anonymity, l-diversity, Laplace differential privacy across four ϵ levels over 8 clinical features).
- Reduced record-linkage re-identification risk from **3.38% to 0.32%** while retaining **99%** of baseline predictive performance.
- Fine-tuned **Qwen3 via 4-bit QLoRA** on an NVIDIA H100 with a curated instruction dataset for controllable long-form generation.

Emerson

Jun 2022 - Apr 2023

ML Trainee

India

- Built supervised regression/classification models for equipment-failure prediction, anomaly detection, and capacity forecasting on operational data.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification, achieving **89% F1** on custom NER tasks.

PROJECTS

Privacy-Preserving Clinical ML Pipeline

Framework combining k-anonymity, l-diversity, and differential privacy to analyze sensitive patient data without exposing identities - quantifying the privacy-utility trade-off of each technique.

MangaLens

Shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation of manga/webtoons across 29+ sites.

career-ops

Autonomous AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025 · GPA 3.70/4.0